

Compositional Linearization for Superposition and Hamiltonianization

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Abstract

test

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hello!

1 Introduction

1.1 Motivation and Overview

Doing stuff is important. $\diamond$ David says: No it's not. $\diamond$ $\diamond$ Maxi says: Yes it is. $\diamond$

2 Monoidal Promonads and Quadratic Forms

2.1 Monoidal Promonads

2.2 Quadratic Forms as a Running Example

2.3 Transporting the Theory of Quadratic Forms

2.4 Algebras for Monoidal Promonads

3 Promonadic Lenses and Para

3.1 Moore Internalization

3.2 Lenses for Monoidal Promonads

3.3 A Reminder on Para

4 Linearization and Critical Points

4.1 The Monoidal Product on Lin

4.2 Retractions and Sections Between Lin and Crit

4.3 A Coalgebraic Lift over Poly

5 Superposition and Hamiltonianization

5.1 Superposition in Physics

5.2 The Poincaré Point of View

5.3 Linearization for Differential Equations